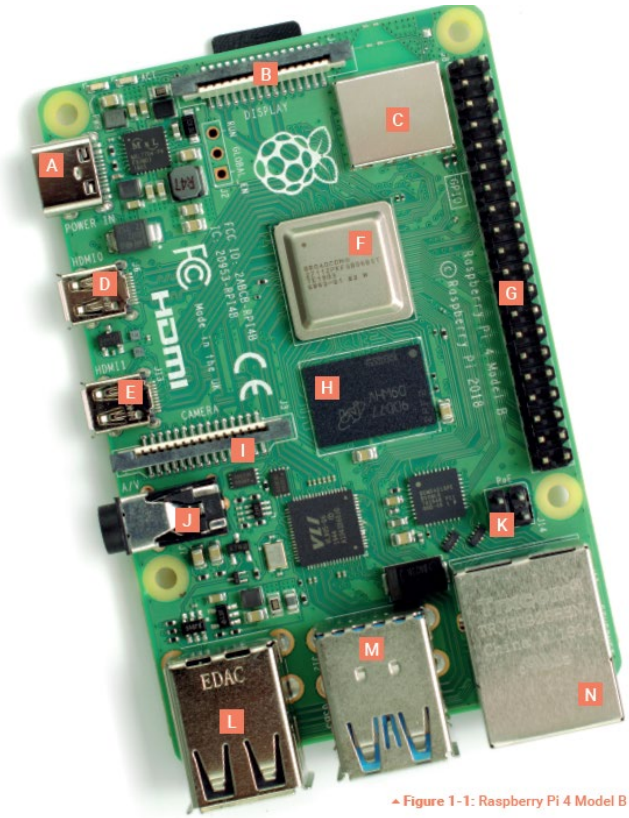
Lab 1: To know and use Raspberry Pi

For your reference, there is an online official E-book prepared to start with Raspberry Pi. (Available here: https://magpi.raspberrypi.com/books/beginners-guide-4th-ed)

This guide is a summary of Chapters 1-3 of the official E-book.

1. To know the hardware

Fig. 1 shows the picture of Raspberry Pi 4B and its hardware components. Refer to Pages 11-19 for a detailed introduction for each component. If there is any doubt, use Google or talk with the professor.



A USB Type-C power in	F System-on-chip	K PoE
B DSI display port	G GPIO	L USB 2.0
C Wireless / Bluetooth	H RAM	M USB 3.0
D Micro-HDMI 0	I CSI camera port	N Ethernet port
E Micro-HDMI 1	J 3.5mm AV	

Fig. 1. Raspberry Pi 4B and hardware functions

2. To preload the Raspberry Pi operating system

For a Raspberry Pi to work, it requires an Operating System (OS). The OS can be preloaded on a microSD card or a USB disk. To do so, we will need the software

Raspberry Pi Imager (See Fig. 2), which can be downloaded from: https://www.raspberrypi.com/software/

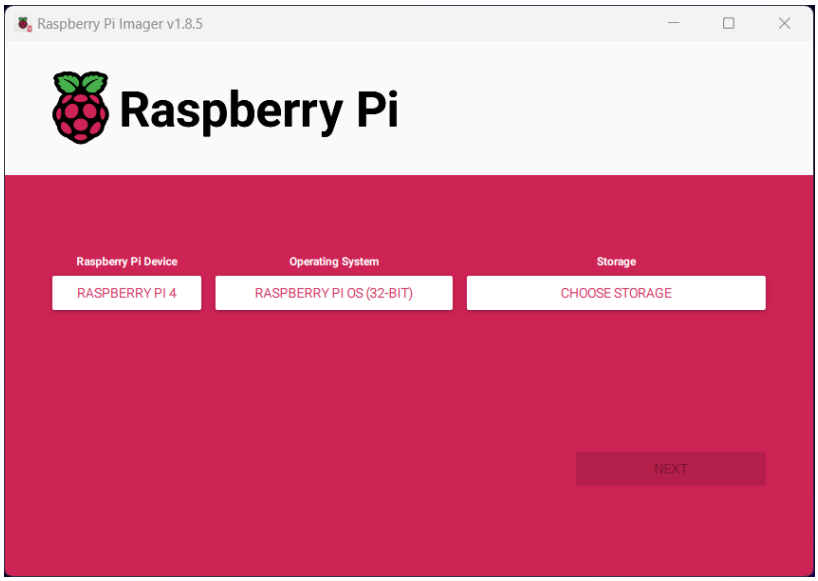


Fig. 2. Raspberry Pi Imager

Once the RPI-Imager is installed successfully, we may set the device, the OS, and the storage to burn the OS. It is recommended to select the 32-bit OS, which has better software compatibility. Depending on the network speed, it may take a while to download the OS and burn it to the storage. Once the preloading is completed, we are ready to set up a computer with Raspberry Pi.

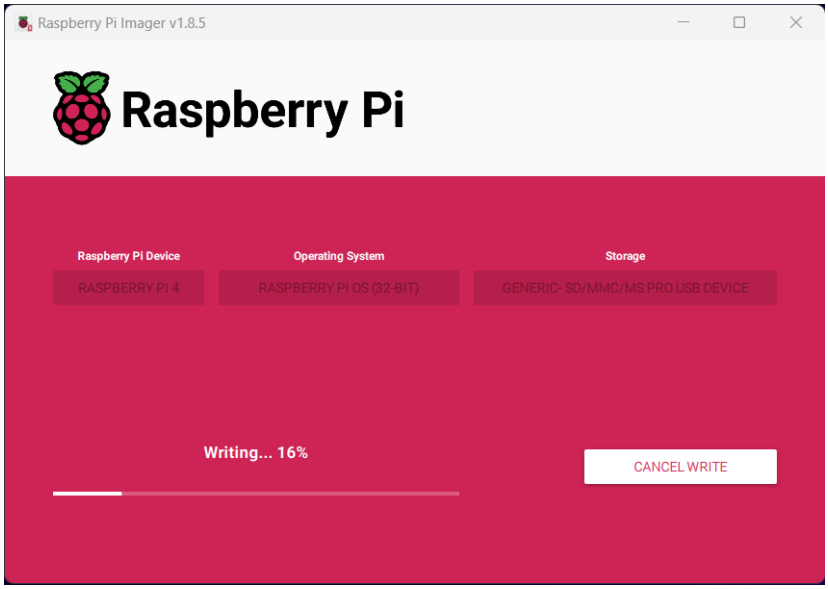


Fig. 3. Preloading the OS to storage

3. To set up a computer with Raspberry Pi

To set up Raspberry Pi, we will need the following:

- One power adapter to supply power;

- One normal PC monitor in the classroom as the display;
- One mini-HDMI cable for connecting the monitor and the Raspberry Pi;
- One USB keyboard and one mouse to interact with Raspberry Pi;
- One microSD card with a preloaded operating system;

Fig. 2. Peripherals to set up a computer with Raspberry Pi

Connect all these devices to make a working computer powered by Raspberry Pi. After the connection is done, plug the power adapter to start it up and wait for a while (the first boot-up might take a bit longer). If everything goes well, you should be able to see the following screen on the monitor. If not, check and fix the problems.

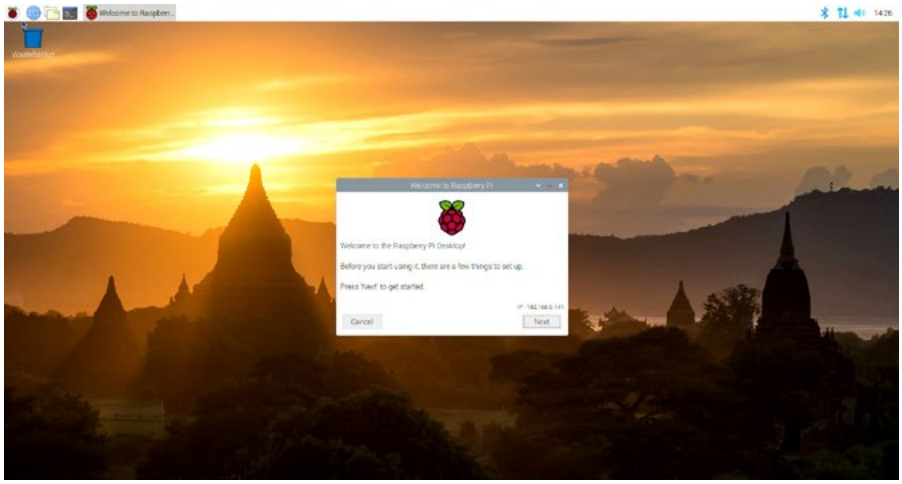


Fig. 3 Screen after a successful start-up

4. To configure and use the OS of Raspberry Pi

For the first-time boot-up, a welcome wizard will pop up for system configuration. Feel free to finish the configuration with your preference. (Recommended to set the language to English, easier for the professor to help afterward ~ ^^)

The Raspberry Pi has a default user and password.

Default user/password:

User: pi
Password: raspberry

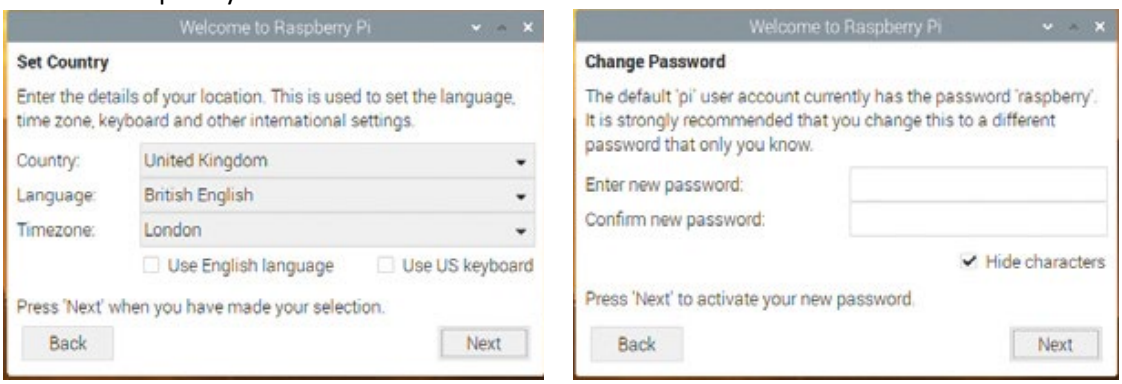


Fig. 4. OS configuration

After the OS setup is ready, we have a computer to play with and it is powered by a Raspberry Pi embedded device. Congrats!

However, it is just a dumb computer and is not so fun. It would be much better if we could connect it to the network. Either we can connect it to the network through the Ethernet port ('N' in Fig. 1) or we can do that with the wireless module ('C' in Fig. 1).

Here, we will do it with the wireless module, specifically the WiFi. You may create a WiFi hotspot using your mobile (preferable with Andriod). Connect the Raspberry Pi (go to the upper-right corner) to the WiFi network you created.

Fig. 5. Network connection

After setting up the network, use the internet browser (upper-left corner) to check if you can visit Google and other websites. If everything goes well, the network is done successfully.

5. File manager and software applications

You may also find that the Raspberry Pi OS has a well-designed file manager and other software applications (as shown in Fig. 6), just like that in the Windows system. Explore them by yourself to understand what this 'small computer' can do.

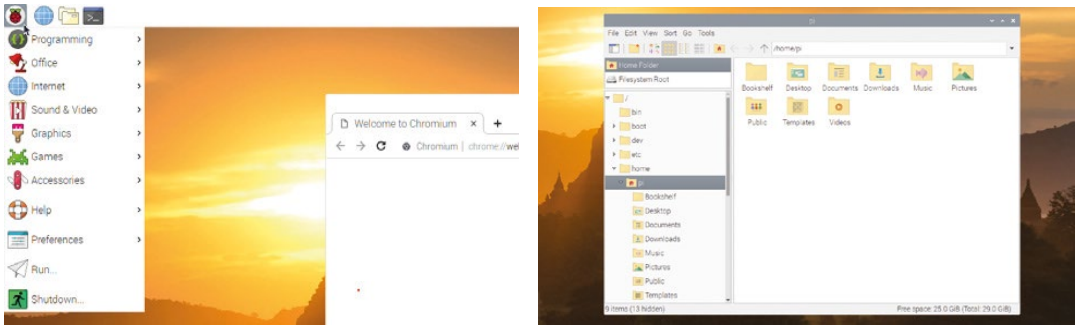


Fig. 6. File manager and software

6. The most powerful tool-Linux Terminal

As you may know, the Raspberry Pi OS is based the Linux, which is a widely used and open-source operating system (if not familiar, take a look at here: https://www.linux.com/what-is-linux/). The most powerful tool loved by the world-wide developers in Linux is its terminal. As a variant of Linux, the Raspberry Pi also has a terminal embedded (the black icon with a '>_' on the upper-left corner). Start the terminal (Fig. 7) and explore how it works.

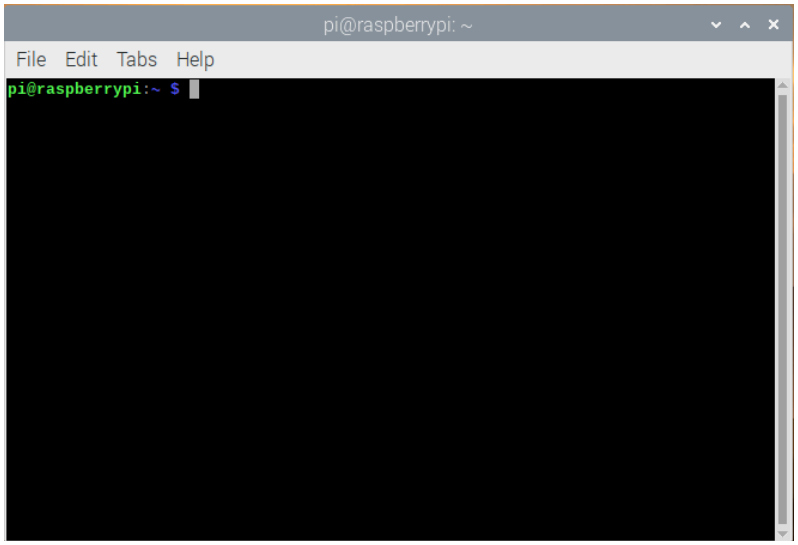


Fig. 7. Terminal of the Raspberry Pi

There are many cheat sheets summarizing the Linux commands that can be used in the terminal. Specific to Raspberry Pi, you can refer to the post here: https://randomnerdtutorials.com/learning-basic-linux-commands-raspberry-pi-cheat-sheet/. (The commands are more or less the same for the OS based on Linux.) The very basic and most important commands to try:

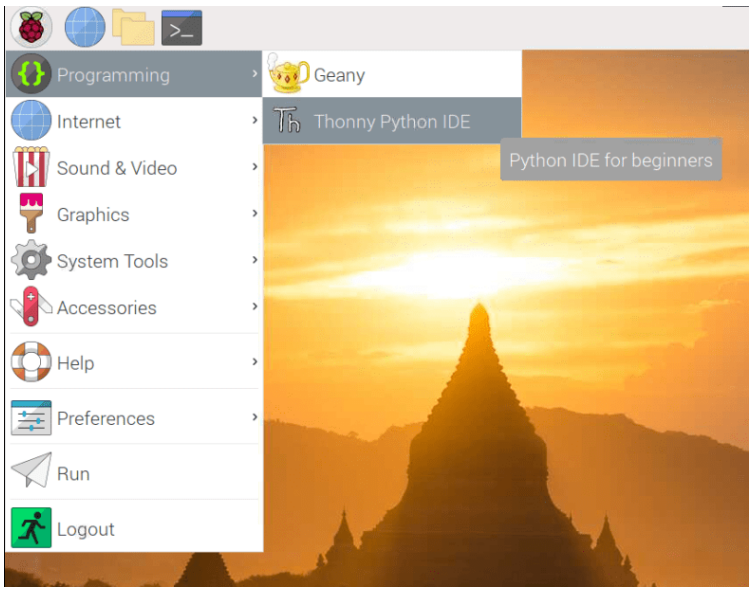
pwd
ls
cd
makdir
cp
mv
rm (a little bit carefulness with this one, it deletes files, folders, etc.)
sudo apt-get update
sudo apt-get install
sudo raspi-config
sudo poweroff
sudo reboot
…

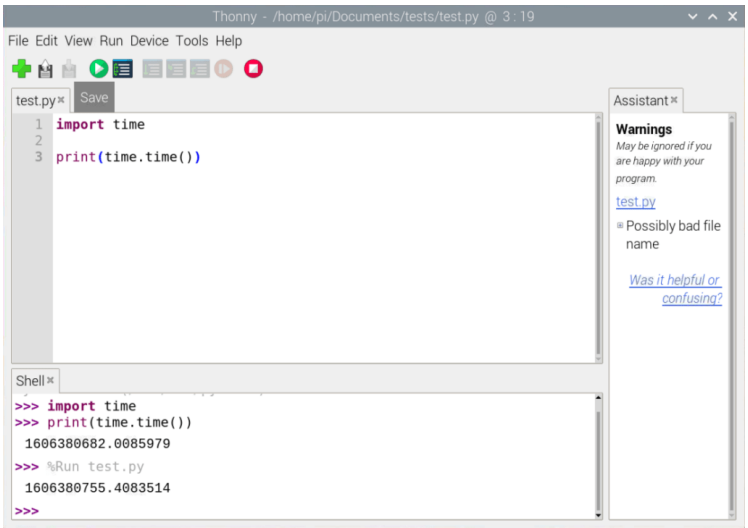
You can also type the command:
python3
nano

7. Python coding with Raspberry PI

On Raspberry Pi OS, it comes pre-installed IDE, Thonny, which offers a simple interface and is easy to use for beginners. It can be found under 'Programming -> Thonny Python IDE -> Python IDE for beginners'. With Thonny, it is straightforward to write and run Python codes.

Advanced users may choose to install VS Code, PyCharm, or lightweight text editors (e.g. nano).




Thonny - /home/pi/Documents/tests/test.py @ 3 : 19
File Edit View Run Device Tools Help
test.py
Save
Assistant
import time
print(time.time())
Warnings
May be ignored if you are happy with your program.
test.py
Possibly bad file name
Was it helpful or confusing?
Shell
>>> import time
>>> print(time.time())
1606380682.0085979
>>> %Run test.py
1606380755.4083514
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